

Text Machine Weekly Status

06/15/2026

Current Status: Power

- Tests with big dynamo:

Cap	Crank Time	Voltage Change	ON Time
50F	1 min	2.6v-3.5v	5 min
50F	30 sec	3.0v-3.5v	5 min
120F	3 min	2.6v-3.4v	10 min
120F	1.5 min	3.0v - 3.4v	10 min

Current Status: Power (cont)

- 50F with big dynamo meets the criteria of 30 second crank time with 5 min text time
- Need to implement an ideal diode MOSFET circuit to isolate the supercap at the lower end of functional voltage (say 3.1v)
- 50F is nice in that, even if supercap drains to 2.6 once in a while, it only takes a minute to get 5 mins of usage
- Solar power has been proven to trickle charge well, and could be added for a larger cap backup power supply, but added complexity and user muddling of power source defeats the patent purpose and ethos of minimizing communication to what the crank delivers
- Option to have a “quick boot” cap and longer crank cap does seem reasonable. The longer crank cap (240F) could be supplementally charged by solar, but I’m not a fan and the added complexity is non-trivial

Current Status: System

- Currently have to type phone number every time and then number disappears while typing
- Before making test run of 5-10 units, GUI/Display needs to have the following capabilities:
 - Contact to message simplicity like standard text app
 - “Time-to-crank” low voltage indicator/alarm
 - Voltage level icon so user can stop cranking and look at the voltage level.
- Minimizing Arduino/Radio/Display power
 - Arduino low power mode enablement
 - Radio sleep time enablement (wake-up on TX send, sleep with 60 sec RX chirps)
 - Screen/touch off when not in use
 - Lower touch refresh rate (example, on initial use a 3-5 sec touch fully enables the display and touch)
- For production run, start board design, using parts of Crankphone board to integrate power (alternatively, we could use old crank lite boards or have the AC/DC rectifier inline soldered to wires).

Current Status: Mechanical

- While the smaller crank and gearbox may work, the K-Tor size dynamo testing has been shown to deliver 30 second crank time.
- Mechanical To-do
 - Build out test unit with K-Tor size dynamo in mind
 - This will probably result in a flat side dynamo handle similar to K-Tor and Eton.
 - Old patent won't apply, but the flat-side handle feels better anyways
 - Source the bigger dynamo for production run (can buy K-Tors for the 5-10 test run)

Current Status: 10 unit build

- Target 8/1/2026